

Time Series Analysis on UFO Sightings

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Abstract—This study analyzes the forecasting abilities between SARIMA and FlowState.

I. INTRODUCTION TO DATASET

The data set contains information about UFO sightings from 1906 to 2014. We use a subset of the data from January 1990 to December 2012 and forecast January 2013 to December 2013. The data is sampled monthly.

II. METHODS

FlowState

FlowState from IBM is a 9.1M parameters model claiming to outperform SOTA TSFM as of July 2025. One of its key innovations is including a scaling parameter and with a state space machine estimating the coefficient for Legendre and Fourier basis functions to reconstruct a continuous signal.

1. Legendre Polynomial Basis (Trend Generation)

FlowState captures non-repeating structural trends using orthogonal Legendre polynomials, formally defined by Rodrigues' formula:

$$P_n(t) = \frac{1}{2^n n!} \frac{d^n}{dt^n} [(t^2 - 1)^n] \quad (1)$$

- $P_n(t)$: The value of the n -th Legendre polynomial at future time t (normalized, e.g., $t \in [-1, 1]$).
- n : The polynomial degree ($n = 0$ is flat, $n = 1$ is linear, $n = 2$ is parabolic, etc.).
- $\frac{1}{2^n n!}$: A scaling constant to bound the curve preventing mathematical explosion.
- $\frac{d^n}{dt^n}$: The n -th derivative, dictating the evolving slope of the underlying algebraic function.

2. Fourier Series (Seasonality Generation)

To reconstruct repeating, cyclical patterns, the architecture utilizes a standard Fourier series of sines and cosines:

$$\text{Seasonality}(t) = \sum_{k=1}^K \left(\theta_{k,s}^{(F)} \sin(2\pi k\omega t) + \theta_{k,c}^{(F)} \cos(2\pi k\omega t) \right) \quad (2)$$

- K : The total number of fixed harmonics, dictating the capacity for complex, high-frequency repeating shapes.
- k : The index of the current harmonic being calculated.
- $\theta_{k,s}^{(F)}, \theta_{k,c}^{(F)}$: The continuous coefficients (weights) predicted by the State Space Model for the sine and cosine components, determining amplitude and phase shift.

- ω : The fundamental base frequency (e.g., daily or weekly cycles), structurally hardcoded prior to training.
- 2π : Constant to convert the linear time variable t into radians.

3. Final Continuous Signal Reconstruction

The final forecast, $\hat{y}(t)$, is the linear combination of the structural trend and repeating seasonality. Because this outputs a continuous mathematical function rather than discrete points, the model achieves complete sampling-rate invariance.

$$\hat{y}(t) = \underbrace{\sum_{n=0}^N \theta_n^{(L)} P_n(t)}_{\text{ML-Scaled Trend}} + \underbrace{\sum_{k=1}^K \left(\theta_{k,s}^{(F)} \sin(2\pi k\omega t) + \theta_{k,c}^{(F)} \cos(2\pi k\omega t) \right)}_{\text{ML-Scaled Seasonality}} \quad (3)$$

- $\theta_n^{(L)}$: The State Space Model's predicted weights for the Legendre polynomials. [1]

SARIMA

The key difference between SARIMA vs ARIMA is in addition to the Auto regressive, Integrated, and Moving Average we have a seasonal parameter as well [2].

Evaluation Metrics

- **RMSE**: Rewards small and consistent errors but penalizes big mistakes.
- **MAE**: Good for detecting how much predictions deviate.
- **MAPE**: Good for detecting how close overall are the predictions.

III. RESULTS AND DISCUSSION

We can see a common point in figure 2.1 were the statistical approach struggles with trend on Figure 1.1. We can prove this by comparing monthly prediction errors between figure 2.1 and 3.1. For SARIMA all of them are negative in contrast to FlowState if the later half they have a positive trend. Lastly, if we set the scale properly to 2 we get lower error scores as seen in figure 4. Figure 1.2 and 1.3 shows our prominent seasonal component alongside with the residual. Figure 1.4 showcases the average counts per month. Figure 1.5 showcases the summer spike in our seasonal component. Figure 1.6 showcases the the high correlation in the values through the years at exactly the 12 month mark. Figure 1.7 backs this with the spike at 12 months.

IV. CONCLUSION

After this study aside from computational efficiency we can observe how FlowState improves upon SARIMA in all aspects, offering more accurate and richer predictions.

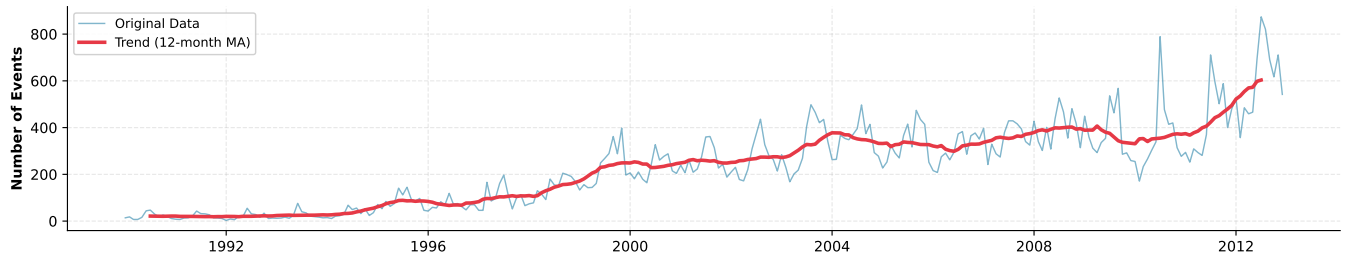
V. REFERENCES

REFERENCES

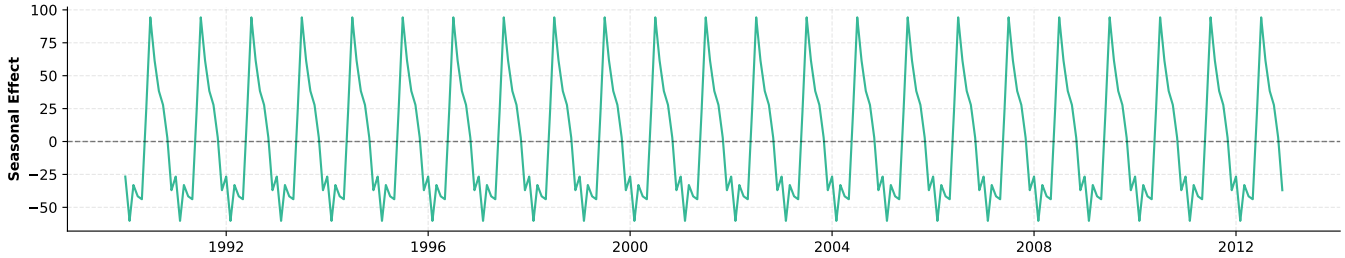
- [1] L. Graf, T. Ortner, S. Woźniak, and A. Pantazi. FlowState: Sampling Rate Invariant Time Series Forecasting. [Online]. Available: <http://arxiv.org/abs/2508.05287>
- [2] A. E. Permanasari, I. Hidayah, and I. A. Bustoni, "SARIMA (Seasonal ARIMA) implementation on time series to forecast the number of Malaria incidence," in *2013 International Conference on Information Technology and Electrical Engineering (ICITEE)*. IEEE, pp. 203–207. [Online]. Available: <https://ieeexplore.ieee.org/document/6676239/>

Comprehensive Seasonality Analysis (Trend + Seasonal + Residual Decomposition)

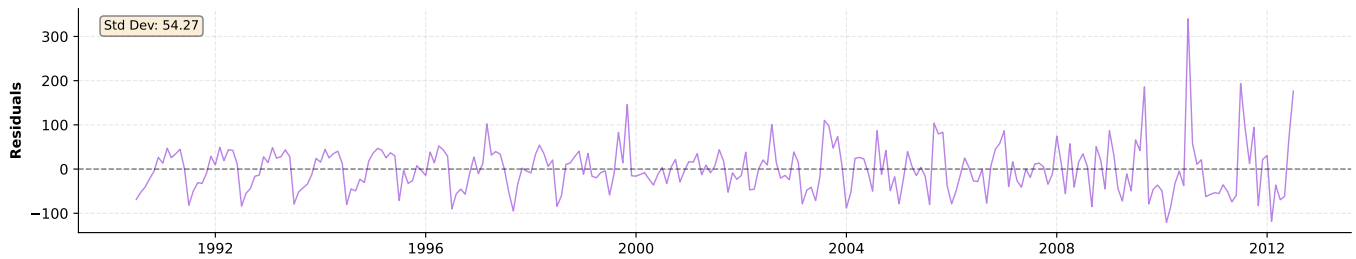
[1] Original Time Series with Trend



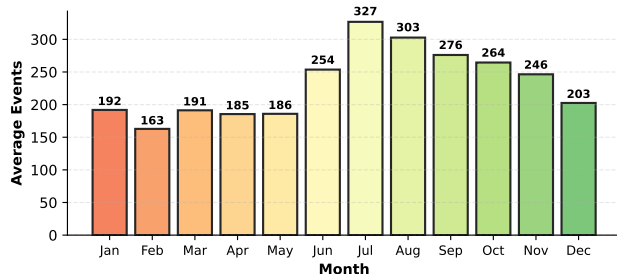
[2] Seasonal Component (Deviations from Trend)



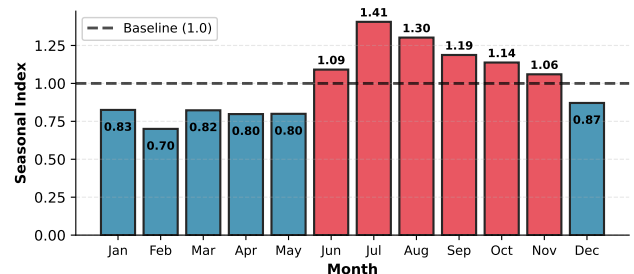
[3] Residual Component (Unexplained Variation)



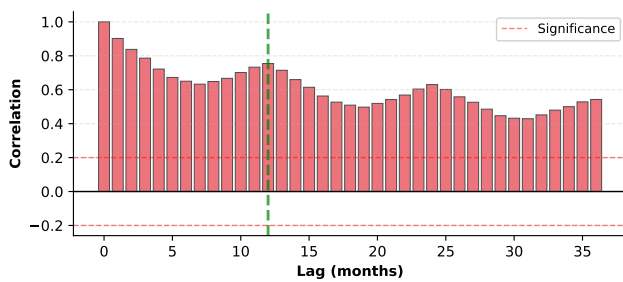
[4] Average Events by Month (Multi-year average)



[5] Seasonal Indices (Multiplicative)



[6] Autocorrelation Function (ACF)



[7] Power Spectrum (Frequency Domain)

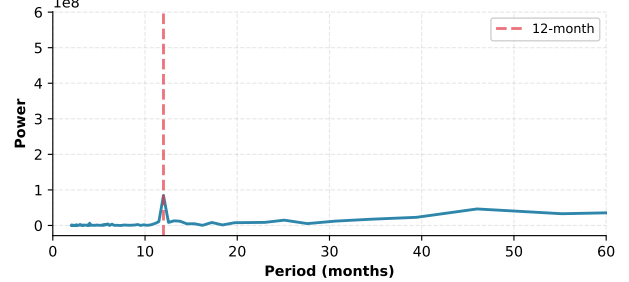


Fig. 1. Graph 1 showcases the trend plotted against the original data with its x axis being time and y the number of events. Graph 2 showcases the seasonal component with y being the seasonal effect. Graph 3 holds our residual component. Graph 4 show the average sightings by month,

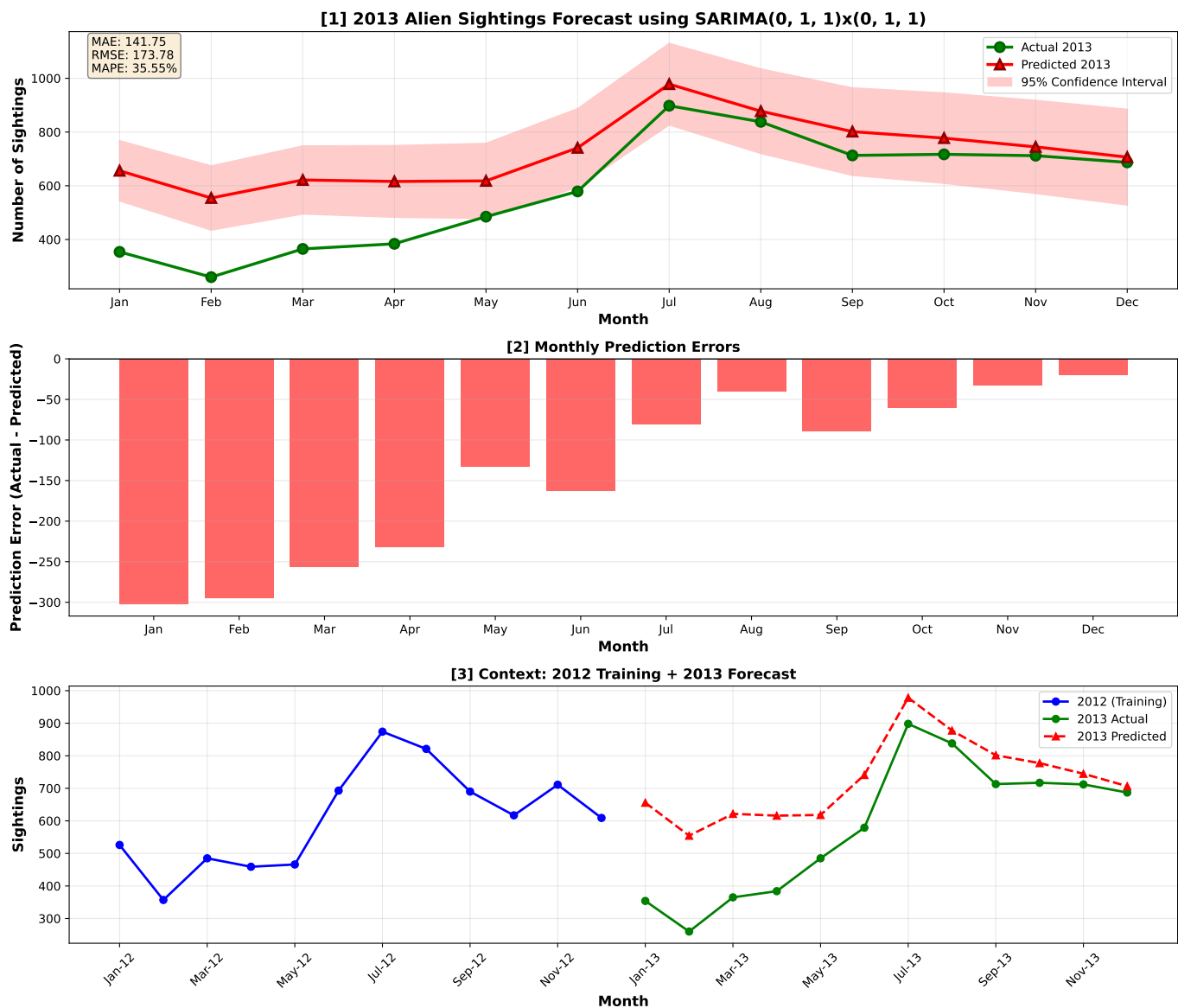


Fig. 2. Graph 1 shows the difference between predicted ARIMA vs actual values with its confidence interval with the top detailing p,d,q x seasonal p,d,q (no auto regressive, First-order differencing, 1 moving average). Graph 1 shows the monthly prediction errors. Graph 3 the previous year.

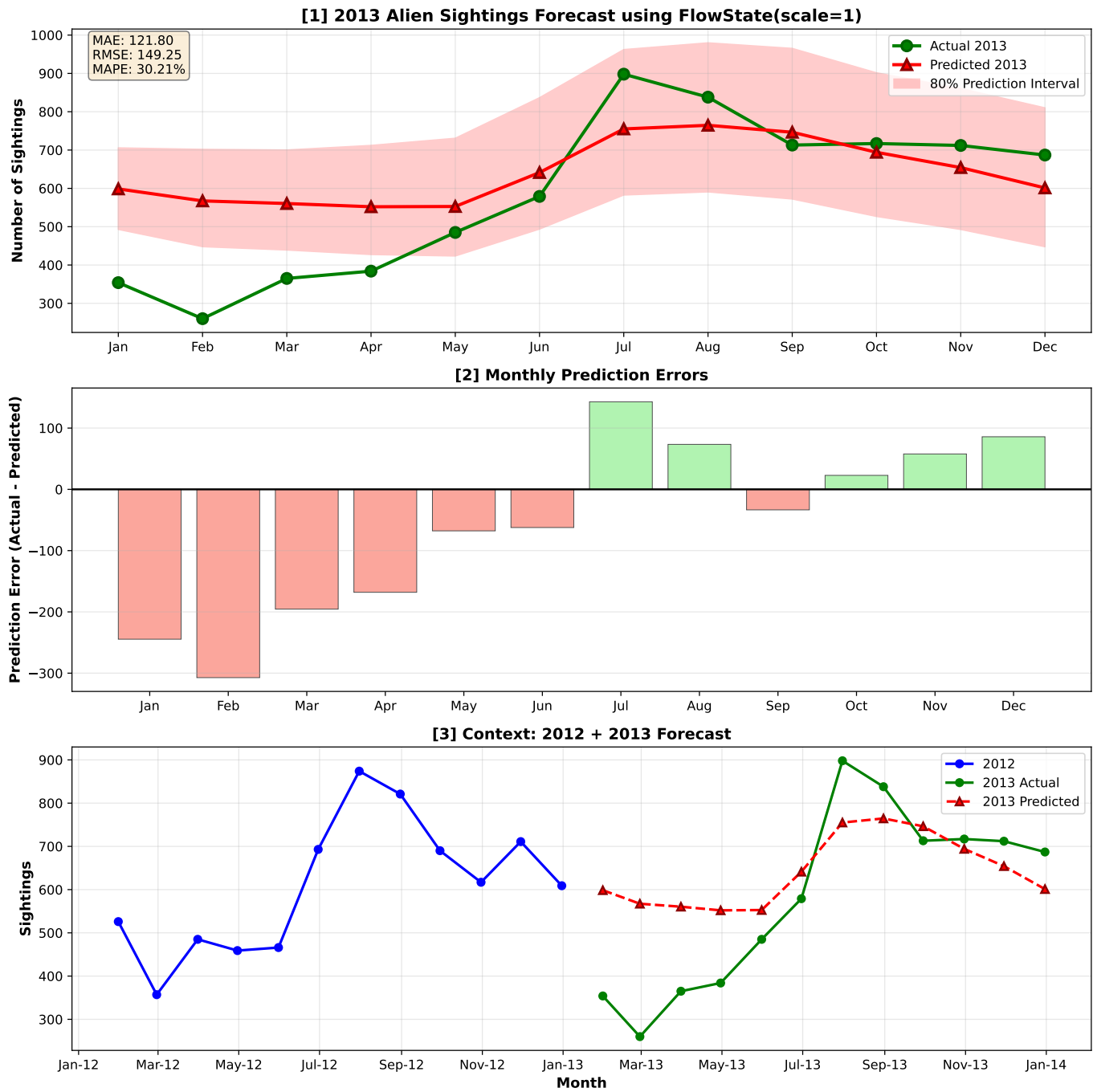


Fig. 3. With the scale parameter set to 1 we the graph 1 showcases the difference between predicted values from FlowState against the actual values with its confidence interval. Graph 2 shows the monthly prediction errors. Graph 3 the previous year and for all the graphs the x axis is time and y is the number of sightings.

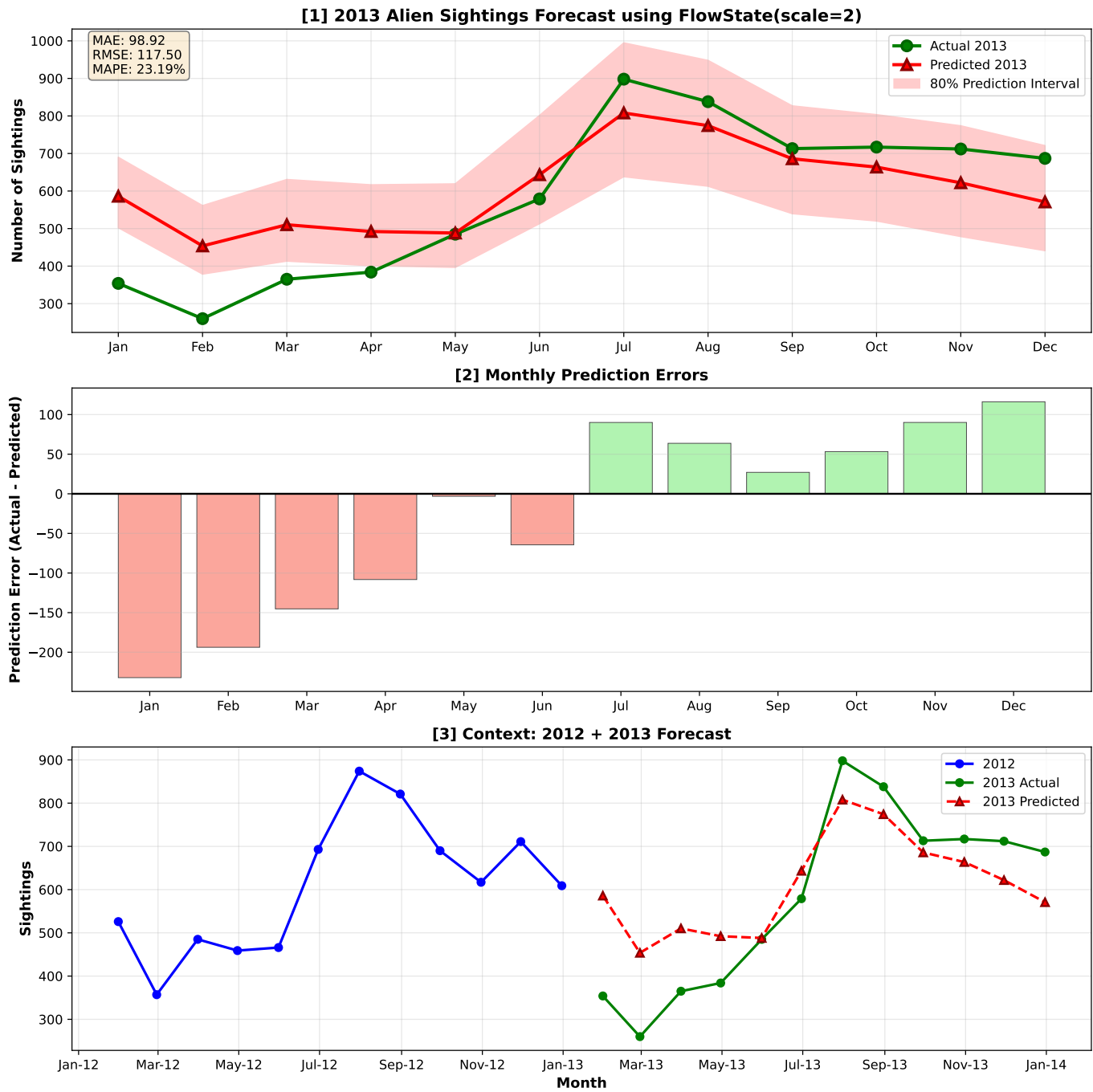


Fig. 4. With the scale parameter set to the suggested 2, graph 1 showcases the difference between predicted values from FlowState against the actual values with its confidence interval. Graph 2 shows the monthly prediction errors. Graph 3 the previous year and for all the graphs the x axis is time and y is the number of sightings.